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Designing a Universal Information Language

Oscar Bender-Stone

March 26, 2026

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Agenda

Introduction

Language Overview

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Background

- Information Management: studies how to **store** and **organize** data

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Problems

- How to *actually* manage information?
  - **Information base:** a database of information
- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]

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- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]
- **Breadth:** different kinds of data
  - Example: Wikipedia[2]
- **Standardization:** ambiguous interpretations
  - Related issue: being **interoperable**
  - **Interoperable:** a portable data format
  - Example: US health records[3]

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Existing Solutions

- **Resource Description Language (RDF)**
  - ▸ Used in Web Ontology Language **(OWL)**
  - \+ Built to be **interoperable**
  - − No built-in logic engine
- **Labeled Property Graphs (LPGs)**
  - ▸ Stores **properties** (node/edge metadata)
  - \+ Efficient queries
  - − Not standardized
- **Cyc**
  - ▸ Largest **knowledge base** to date
  - \+ Sophisticated logic engine
  - − Hard-coded, proprietary rules

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Proposed Solution

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- **Welkin**: an information language

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  - Old German word, meaning “cloud”, “sky”

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- **Goals:**
  - **Universality:** be completely expressive

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    - Describe improvements *in* Welkin

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  - **Standardization:** be completely *unambiguous*
    - Have a formal specification

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Base Operations

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